mmWave-Based Contactless BP Monitoring With Physio-Model-Guided Deep Learning

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***Abstract*—Blood pressure (BP) is a critical indicator for life-threatening conditions. While invasive catheter-based methods offer high accuracy, non-invasive techniques typically require placement on specific body areas, introducing discomfort and rendering their accuracy sensitive to wearing conditions. To overcome these limitations, recent efforts have explored contactless BP monitoring using RF sensing. However, existing approaches often rely on deep learning models without grounding in physiological principles, resulting in poor generalization and limited clinical trustworthiness. In this paper, we propose *h*BP-Fi, a contactless BP measurement system driven by *hemodynamics* acquired via RF sensing. In addition to its contactless convenience, *h*BP-Fi outperforms existing RF-based approaches by i) employing a physiologically grounded hemodynamic model of pulse generation that forms the basis for RF-based BP estimation, ii) enabling super-resolution arterial pulse tracking via beam-steerable RF scanning, iii) ensuring output trustworthiness through an interpretable (transparent-by-design) deep learning model, and iv) achieving robust generalizability to unseen users and scenarios via a CycleGAN-based training strategy. Extensive experiments with 35 subjects under practical scenarios demonstrate that *h*BP-Fi can achieve errors of -2.95 ± 7.66 mmHg and 2.63 ± 6.05 mmHg for systolic and diastolic blood pressures, respectively.**

***Index Terms*—Blood pressure, mmWave radar, hemodynamics, neural network.**

I. Introduction

HYPERTENSION (or high blood pressure) is the leading risk factor for mortality worldwide. It affects nearly half of all adults in the United States [1] and more than 1.3 billion people globally [2]. Despite its prevalence, hypertension often develops silently over many years without noticeable symptoms, eventually leading to severe health conditions such as heart attacks, strokes, chronic heart failure, and kidney disease [1]. Therefore, it is critical to monitor *blood pressure* (BP) regularly to achieve timely diagnoses. In out-of-clinic settings, regular BP measurements are typically non-invasive, relying on inflatable cuffs or sensor-based devices placed on the arm [3], [4], chest [5], [6], wrist [7], [8], and ears [9], [10]. However, these methods require firm contact at specific body sites, which may cause discomfort and limit use in individuals with sensitive or injured skin. Moreover, the accuracy of such contact-based measurements is highly dependent on proper device placement and stable wearing conditions—factors that are often difficult to maintain in daily use, which can hinder their practicality for long-term or continuous monitoring.

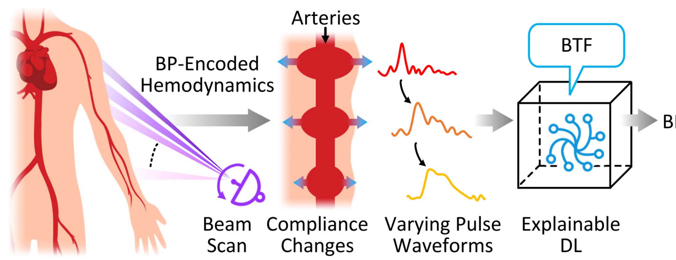

Fig. 1. The conceptual construction of *h*BP-Fi: from hemodynamics to RF micromotion sensing and interpretable DL inference.

Recent progress in contactless sensing [11], [12], [13], [14], [15] brings new hope to alleviate these limitations. Among these, *radio-frequency* (RF) sensing devices with sufficient bandwidth and spatial resolution are a promising direction. Existing systems [16], [17], [18] sense chest/arm motions as external biomarkers and map waveform morphological features to BP via *deep learning* (DL). However, because the learned mapping lacks grounding in physiological mechanisms, its causal validity is difficult to establish. DL models' interpretability issues in network design and output decisions further constrain deployment in critical medical settings. Additionally, these systems demand large training volumes to handle morphological variations across subjects and scenarios. Consequently, we need approaches grounded in sound physiology, with efficient training and better interpretability.

To this end, we present *h*BP-Fi to associate, for the first time, advanced RF sensing and DL inference with a sound physiological basis; it leverages motion-sensitive RF signals to capture the BP-encoded hemodynamics of pulses transiting along arm arteries. As shown in Fig. 1, as blood flows

Received 5 November 2024; revised 18 November 2025; accepted 13 December 2025. Date of publication 24 December 2025; date of current version 7 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62372045 and Grant 62502274 and in part by Shandong Provincial Natural Science Foundation under Grant ZR2025QC1569. Recommended for acceptance by J. Widmer. *(Corresponding authors: Fan Li; Jun Luo.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Shandong University Ethics Review Committee.

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Digital Object Identifier 10.1109/TMC.2025.3647930

TABLE I
COMPARATIVE OVERVIEW OF NON-INVASIVE BP MEASUREMENT METHODS. (A COMPREHENSIVE REVIEW IS PROVIDED IN SECTION II)

Work	eBP [9]	Glabella [10]	Crisp-BP [7]	Fan *et al.* [19]	Ebrahim*et al.* [20]	mmBP [17]	*h*BP-Fi
Sensing modality	Oscillatory	PPG	PPG	Vision	RF	RF	RF
Non-physical contact	×	×	×	✓	×	✓	✓
Void of extra hardware	×	×	✓	✓	×	✓	✓
Output interpretability	✓	×	×	×	×	×	✓
Domain adaptability	×	×	×	×	×	×	✓
Physiological basis	✓	✓	✓	✓	×	×	✓
SBP mean/std (mmHg)	1.8/7.2	±10	1.67/7.31	8.42/8.81	<10	0.46/4.87	-2.05/6.83
DBP mean/std (mmHg)	-3.1/7.9	N/A	0.86/6.55	12.34/7.10	N/A	1.13/5.14	1.99/6.30

through, the arm arteries undergo compliance changes indicating changes in BP [21], resulting in variable pulse waveforms depending on the measurement site [22]. Consequently, the varying pulse waveforms measured at different sites allow for deriving a *BP-specific transfer function* (BTF); inferring BP becomes possible as the BTF, describing how compliance varies along arteries, is causally related to BP. Specifically, *h*BP-Fi carefully steers RF beams to scan an arm, aiming to capture varying pulse waveforms along the arm. It then incorporates an interpretable DL pipeline to estimate BP values, where the pipeline's architecture emulates the BTF derivation process, and the outputs are constrained by BTF-related terms. As demonstrated in Table I, these innovations enable *h*BP-Fi to overcome key limitations of contemporary non-invasive BP measurement technologies: (a) superior generalizability by avoiding overfitting to dataset-specific artifacts, (b) enhanced robustness through physiologically-constrained learning, and (c) network interpretability missing in pure DL solutions.

Achieving the vision of *h*BP-Fi involves several critical challenges. First, the interaction between RF signals and hemodynamic processes remains insufficiently characterized for BP estimation. Second, accurate hemodynamic profiling requires capturing high-quality pulse waveforms from multiple arterial sites along the arm. Yet, despite the beam-steering capabilities of advanced RF devices (e.g., mmWave radar [23]), the spatial resolution is insufficient to isolate individual locations due to signal mixing from neighboring tissues, making direct hemodynamic profiling with beam steering alone impractical. Third, environmental factors such as time and temperature fluctuations can significantly affect BP estimation based on BTF. While DL offers a path to robust performance under such variability, developing interpretable models that gain trust in clinical settings remains an open research problem, particularly in clinical-facing applications. Finally, pulse waveforms acquired via RF sensing often vary significantly across users and scenarios, leading to domain shifts that can degrade inference accuracy. A practical solution must therefore generalize well to unseen conditions without requiring domain-specific calibration.

To address these challenges, we first carefully study the hemodynamics along arm arteries to figure out how the changes of RF-sensed pulse waveforms along the arm encode BP information. In particular, we characterize the relation between BP and varying pulse waveforms as a modified tube-load model [24], which captures the underlying physiological dynamics. To acquire these waveforms, we then specifically design a continuous, beam-steerable RF sensing scheme equipped with inference detection. Although each individual scan cannot isolate a specific arterial site due to spatial diffusion, our super-resolution strategy aggregates information across the entire scanning sequence to reconstruct high-fidelity pulse waveforms along the arm. Finally, we develop a CycleGAN-based DL pipeline to process the extracted waveforms. By embedding the BTF-based physiological inference process into the network architecture and training, our method achieves strong cross-domain generalizability while maintaining interpretability—enabling accurate BP estimation across diverse users and conditions. We highlight our main contributions as follows:

- Built upon a solid physiological basis, we propose *h*BP-Fi as the first contactless BP inference system leveraging interpretable deep-analyzed hemodynamics.
- Through a thorough study of hemodynamics, we uncover the intrinsic link between arterial compliance and BP variation. This motivates us to introduce BTF to characterize this relation for the first time, whose parameters are specifically defined to be amenable to beam-steerable RF sensing.
- We propose a continuous beam scan RF sensing scheme that achieves super-resolution pulse waveform recovery (finer than minimum angular separation); it incorporates interference detection to ensure high-quality signal acquisition, and enables multi-point hemodynamic measurement along the arm.
- We develop a physiology-constrained DL pipeline for BP inference, where the model architecture and loss function are explicitly designed using BTF principles and the decision process maintains medical interpretability while achieving clinical-grade accuracy.
- To enhance generalizability, we further improve the BP inference model using a CycleGAN-based domain adaptation framework, enabling reliable BP inference for new subjects and scenarios without calibration.
- We thoroughly evaluate *h*BP-Fi via extensive experiments. The results demonstrate that *h*BP-Fi has an average error of -2.95 ± 7.66 mmHg and 2.63 ± 6.50 mmHg for *systolic blood pressure* (SBP) and *diastolic blood pressure* (DBP), respectively, which is within the acceptable range regulated by the FDA's AAMI protocol [25].

The paper is organized as follows: Section II introduces existing BP measurement methods and motivates our design. Section III presents the hemodynamics modeling and system design of *h*BP-Fi. Section IV reports the evaluation results.

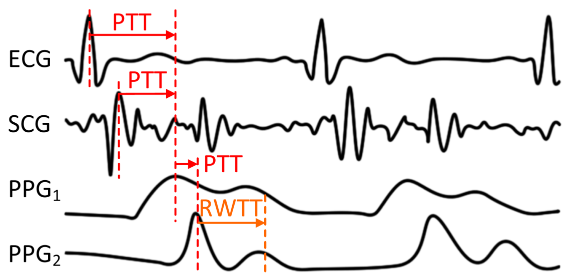


Fig. 2. Wearable sensors measure PTT and RWTT based on physiological signals obtained from multiple sites. Curves are adapted from [7], [10], [28], [29].

Section V discusses limitations and potentials, before concluding in Section VI.

II. Background and Motivations

In this section, we review the existing BP measurement approaches and discuss some of their major limitations, providing context to motivate our design of *h*BP-Fi.

A. Existing Contact-Based BP Measurement

While intravascular catheterization is the gold standard for BP [26], oscillometry [3], [4] is the most common solution due to its accessibility and noninvasiveness. Its normal procedure involves an inflatable arm cuff to compress vessels and determine SBP and DBP based on the applied pressure levels. However, such cuff-based solutions have been widely criticized for causing discomfort by blocking the blood flow and for having system performance sensitive to cuff type [27]. Although eBP [9] and HeartGuide [8] attempted to reduce discomfort by integrating inflatable cuffs into earbuds and smartwatches, their measurement performances are still sensitive to device-wearing states, thus hindering reliable BP monitoring.

With the advances in wearables and mobile devices, later efforts have leveraged easily accessible physiological signals acquired by these devices to infer BP. They are mainly based on the famous physiological fact that *pulse wave velocity* (PWV) is intrinsically related to BP variation [30], by PWV $= \ell/\text{PTT}$ where PTT refers to the *pulse transit time*, the traveling time of pulse waves between two arterial sites, and ℓ denotes the distance in-between; here *pulse* is a shortened form of *blood volume pulse* [31]. This actually converts BP measurements to BP inference via PTT measurements. Fig. 2 shows examples of measuring PTT from peak delay between two physiological signal pairs, such as ECG-PPG [6], [28], SCG-PPG [29], [32], and PPG-PPG [10]. However, these approaches are both inconvenient and unreliable, because they need to attach multiple wearable sensors to a body and their performances are highly dependent on sensor placements: tiny changes on the sites of any sensors can cause dramatic performance degradation [17]. Recently, Crisp-BP [7] exploits *reflected wave transit time* (RWTT) readily obtainable by a wrist PPG sensor to replace PTT for inferring BP, yet it is still sensitive to wearing states due to its wearable nature.

B. How About Contactless Solutions?

Given the drawbacks of contact-based solutions, such as inconvenience, discomfort for some individuals, and sensitivity to device-wearing states and placements, it is natural to consider resorting to contactless solutions. We first briefly discuss a few typical attempts, and then turn our focus to the major obstacle preventing them from getting practical.

1) Why Existing Contactless Fails: Early attempts centered on using cameras to deduce PTT from PPG and in turn infer BP [19], [33], [34]. However, camera-based solutions require strict recording conditions, such as adequate lighting and the entire view of a face. Besides, the incurred privacy concerns further limit their wide adoption. Leveraging RF-sensing to obtain PTT has been attempted in [20], [35], where PTTs are derived from measurements taken by both RF sensing and ECG/PPG sensors. While precisely synchronizing these sensors (of very different modalities) can be extremely challenging, the coarse-grained RF sensing fails to remain robust to PTT's high sensitivity to subtle changes in the measurement site [17]. All these eventually yield inaccurate BP estimations. The same reason also prevents RF-based RWTT measurement from being a realistic BP indicator [36].

While physiologically-grounded approaches grapple with the aforementioned challenges, recent years have witnessed a surge in RF-based contactless BP solutions that adopt a purely data-driven, *black box* DL pipeline [16], [17], [18], [37]. These approaches leverage RF signals to capture pulse waveforms and use deep neural networks to directly map morphological features to BP values, with some reports achieving clinical-grade accuracy (e.g., MAE$<$5 mmHg) in constrained settings. This reported performance, however, contrasts sharply with the method's lack of physiological plausibility. The derived mapping is correlational and non-causal, offering no principled rationale for its predictions. Without a physiological basis or interpretability, the models' reliability is questionable, as they are prone to overfitting spurious correlations in the training data, leading to biased estimates on unseen data [38]. Moreover, the generalization of these models is inherently limited by their training datasets. They often fail to capture the full biological variability of pulse signals across individuals, time, and situations [39], [40]. Consequently, while accurate in controlled experiments, their performance likely degrades in diverse daily-life scenarios, hindering practical clinical adoption.

2) Acquiring Pulse Activities by Scan: Recent advances in RF sensing have equipped devices (e.g., mmWave radar [41]) with antenna arrays capable of steering radiation beams towards specific directions. This capability naturally raises a critical question: *Can beam-steerable RF sensing enable PTT estimation by extracting pulse waveforms from two distinct directions using a single device, thereby overcoming the limitations of existing RF-based PTT methods that require additional sensing modalities and cumbersome multi-modal synchronization?* Unfortunately, our investigation reveals that this approach is fundamentally infeasible due to two intrinsic limitations: (i) Insufficient Angle Resolution: Conventional beam-steering relies on spatial diversity (e.g., Tx-Rx antenna pairs) for angular

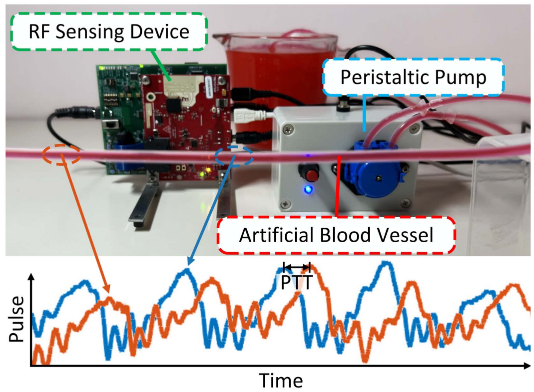
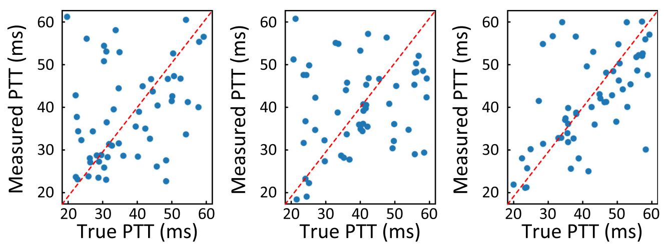


Fig. 3. In-vitro blood circulation simulator to study PTT measured by RF sensing technology.



(a) 4 Tx-Rx pairs (b) 6 Tx-Rx pairs (c) 8 Tx-Rx pairs

Fig. 4. Comparing measured PTTs with true PTTs given different numbers of Tx-Rx pairs.

resolution, but the resulting time-delay precision proves inadequate for PTT estimation. (ii) Improper Biophysical Modeling: The measured pulse signal represents a superposition of reflections from multiple nearby locations, making it unsuitable for position-sensitive PTT measurement methods.

To empirically validate these limitations, we build a blood circulation simulator (Fig. 3); it uses a peristaltic pump [42] to emulate blood flow through vessels at different PWVs. [1] A mmWave radar [23] is adopted to perform beam-steerable sensing. We in particular pick two measurement sites and measure the time delay between the pulse waveforms extracted from these two sites (elaborations on how to obtain pulse waveforms are given in Section III-B2). The true PTT is computed as the ratio of the distance between the measurement sites and the known PWV specified by the pump. As shown in Fig. 4, the results exhibit significant deviations from ground-truth PTT values (error persistent even when increasing the number of Tx-Rx pairs to enhance angular resolution). This confirms that merely repurposing beam-steering for PTT estimation is inherently limited: angle resolution alone cannot compensate for the absence of a biophysical model linking wave dynamics to BP.

Motivated by the above observations, we address these challenges through two key innovations: (i) establish a new biophysical model that reveals the link between pulse activities and BP in Section III-A, and (ii) design an effective beam-steerable sensing scheme for achieving super-resolution extraction of pulse waveforms in Section III-B.

[1] Since normal PWV is 6.84 ± 1.65 m/s [43], we control the PWVs to remain in the range of 3 m/s to 9 m/s by adjusting the pump.

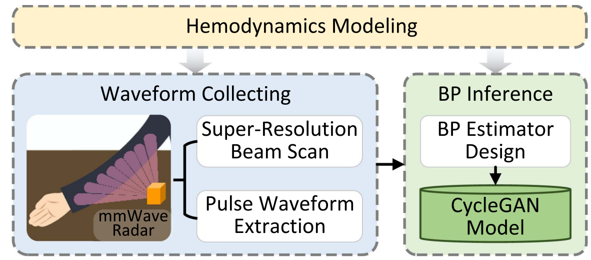


Fig. 5. System architecture of *h*BP-Fi.

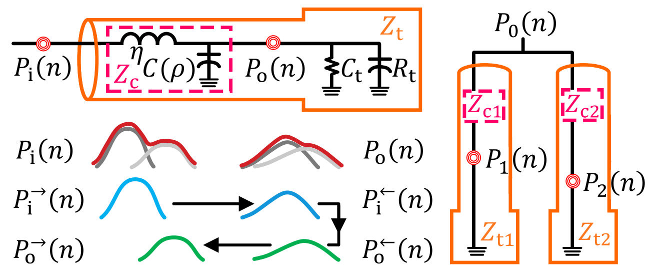
III. System Design

As illustrated in Fig. 5, *h*BP-Fi is mainly composed of three components: *Hemodynamics Modeling*, *Waveform Collecting*, and *BP Inference*. Since vascular compliance (elastic modulus) is affected by BP, *Hemodynamics Modeling* aims to associate the pulse activity captured by RF sensing with the implicit compliance property. To this end, a tube-load model is adopted to describe the hemodynamics, which leads to a novel mapping, BTF, that relates changes in the compliance to those in the pulse waveforms propagating through different measurement sites. With this theoretical basis, *h*BP-Fi leverages an mmWave radar in *Waveform Collecting* to achieve adequate resolution for pulse activity sensing by extracting pulse waveforms out of phase changes in RF reflections along the arm, and identifies the presence of interference through periodicity examination. Finally, to cope with the uncertainties brought by unseen subjects and varying scenarios, *BP Inference* of *h*BP-Fi employs an interpretable DL pipeline, based on CycleGAN structure, to reliably infer BP across various domains. For the actual measurement (also shown in Fig. 5), we require the subject to i) stretch his/her arm on a table, ii) maintain minimal body movement, iii) keep a fixed distance (e.g., 50 cm) from the radar (on the same table) whose field of view (FoV) sufficiently covers the whole arm.

A. Hemodynamics Modeling Via BTF

Hemodynamics refers to the process of how blood flows through the blood vessels. As the heart beats, it pumps blood through the circulatory system formed by blood vessels, generating pressure to push the elastic vessel wall outwards, which is then bounced back after the blood passes through. Among all physiological indicators of hemodynamics, BP indicates the radial pressure caused by the blood flow, whose continuous representation is pulse activity.

Herein, we explain the hemodynamics based on the widely recognized tube-load model [24] with additional accounting for BP-dependent arterial compliance and peripheral wave reflection. While real arteries exhibit complex viscoelastic properties and branching structures, we adopt a simplified representation, as shown in Fig. 6(a), where a vessel segment is modeled as a uniform tube with characteristic impedance $Z_c = \sqrt{\eta/C(\rho)}$. Here, η represents blood inertance, $C(\rho)$ is the BP-dependent



(a) The tube-load model of blood vessel. (b) An instance.

Fig. 6. Hemodynamics modeling (a) and an instance of modeling BTF at different sites (b).

arterial compliance, and ρ denotes BP. This simplification is justified because: : (i) the dominant wave propagation effects are preserved, and (ii) the model parameters can effectively capture the aggregate behavior of more complex systems, as demonstrated in prior studies [21], [44], [45]. According to the Young's modulus [46], the relationship between compliance of the elastic tube and the distending pressure ρ follows an exponential expression [21]:

$$C(\rho) = C_0 e^{-\alpha\rho}, \tag{1}$$

where α is the vessel parameter and C_0 is Young's modulus for zero arterial pressure. Their values are constant and subject-specific, as widely validated by existing studies [47], [48]. Besides, the peripheral vessels are modeled as the three-element Windkessel terminal load [49], which has an impedance Z_t determined by i) characteristic impedance Z_c, ii) peripheral resistance exerted by the arterioles R_t, and iii) compliance of the distal arteries C_t, via $Z_t = Z_c + R_{t/}(1 + R_t C_t)$.

We denote the pulse waves at the tube inlet as $P_i(t)$ and at the outlet as $P_o(t)$. They consist of forward and backward waves, both traveling with delay and distortion characterized by a wave propagation function $\phi(\rho)$ via $P_o^{\rightarrow}(t) = e^{\phi(\rho)\ell} P_i^{\rightarrow}(t)$ and $P_i^{\leftarrow}(t) = e^{\phi(\rho)\ell} P_o^{\leftarrow}(t)$ [24], where the superscripts $\rightarrow$ and $\leftarrow$ represent the forward and backward components, respectively, and ℓ is the length of the tube. Moreover, the backward waves are generated by the forward ones encountering a change in impedance at the terminal load, leading to the following relation:

$$P_o^{\leftarrow}(t) = \Psi P_o^{\rightarrow}(t) = \frac{as}{s^2 + bs} P_o^{\rightarrow}(t), \tag{2}$$

where s is the Laplace variable, $a = 1/2Z_c C_t$, and $b = a + 1/R_t C_t$. Using (2) and based on that $P_o(t)$ and $P_i(t)$ are the sum of forward and backward waves, the relation between $P_o(t)$ and $P_i(t)$ can be modeled by a *BP-specific transfer function* (BTF) $\Gamma_{i\rightarrow o}$ as:

$$P_o(t) = \Gamma_{i\rightarrow o} P_i(t) = \frac{1 + \Psi}{e^{\phi(\rho)\ell} + e^{-\phi(\rho)\ell}\Psi} P_i(t). \tag{3}$$

combining wave transmission ($e^{\pm\phi\ell}$ terms) and reflection (Ψ term). Considering that the tube is frictionless, $\phi(\rho)$ can be reduced to $\sqrt{\eta C(\rho)}$ via the Taylor series expansion [50], yielding the final BTF form:

$$\Gamma_{i\rightarrow o} = \frac{s^2 + (b + a)s}{(s^2 + bs)e^{\ell\sqrt{\eta C(\rho)}} + ase^{-\ell\sqrt{\eta C(\rho)}}}, \tag{4}$$

where polynomial coefficients a, b, and constants for $C(\rho)$ (i.e., η, C_0, α) can be initialized using population-based normative values [51] for efficiency, then refined via subject-specific optimization or DL. Thus, the BTF depends only on ℓ and ρ. Ideally, given observed $P_o(t)$ and $P_i(t)$, one can first measure the BTF via $\Gamma_{i\rightarrow o} = P_o(t) P_i^{-1}(t)$. Then the BP value ρ can be determined by (4) in the least-squares sense, with prior knowledge of the precise artery length between the two sites, i.e., ℓ. It is worth noting that the above relation still holds when $P_i(t)$ and $P_o(t)$ are superimposed by multiple pulses, which is equivalent to the parallel connection of multiple tubes with loads [52].

In practice, precise measurement of ℓ is the major obstacle to solving BTF and inferring BP. Rather than perfecting the estimation of ℓ, we relate BP and BTF via optimization. We consider three pulse waveforms measured at different locations (Fig. 6(b)), denoted by $P_0(t)$, $P_1(t)$, and $P_2(t)$. According to Eqn.(3), we have $P_1(t) = \Gamma_{0\rightarrow 1}(\Theta)P_0(t)$ and $P_2(t) = \Gamma_{0\rightarrow 2}(\Theta)P_0(t)$, where $\Theta \triangleq \{\rho, \ell_{0\rightarrow 1}, \ell_{0\rightarrow 2}\}$ defines the parameter set. Since $\Gamma_{0\rightarrow 1}(\Theta)^{-1}P_1(t) = P_0(t) = \Gamma_{0\rightarrow 2}^{-1}P_2(t)$, we tune Θ by minimizing:

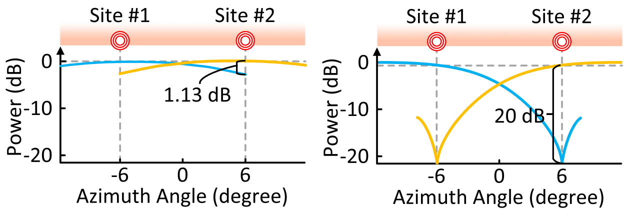
$$\min_{\Theta} J(\Theta) = \min_{\Theta} \left\| \Gamma_{0\rightarrow 2}(\Theta)^{-1}P_2(t) - \Gamma_{0\rightarrow 1}(\Theta)^{-1}P_1(t) \right\|_1 . \tag{5}$$

Several mathematical relations can be incorporated to narrow the searching space for the optimal Θ. Under our context, RF sensing may help infer the radial range and bearing between the RF device and the measurement sites [53], which in turn leads to estimations of $\ell_{0\rightarrow 1}$, $\ell_{0\rightarrow 2}$, and $\ell_{1\rightarrow 2}$ via trigonometric operations. More importantly, (5) can be extended to accommodate an arbitrary number n of pulse waveforms measured at different locations, further helping constrain the parameter set $\Theta \triangleq \{\rho, \ell_{0\rightarrow 1}, \ell_{0\rightarrow 2}, \ldots, \ell_{0\rightarrow n}\}$ and potentially yielding a more accurate estimation to BP ρ. In the following, we shall design *h*BP-Fi to capture fine-grained pulse waveforms from as many well-distinguishable skin sites as possible, so as to achieve accurate BP inference.

B. RF Pulse Waveform Collection

*h*BP-Fi employs steerable RF beams to scan the skin vibrations induced by the pulse along human arm. Particularly, steerable beam patterns are employed in a time-division manner, enabling the RF beam to scan successive directions with a μs-level delay to capture spatially separated pulse waveforms along an arm. In the following, we first discuss how to achieve a super-resolution beam scan, then we elaborate on the waveform extraction scheme and interference detection method.

1) Super-Resolution Beam Scan: RF signals transmitted and received by mmWave radars (with MIMO antennas) are often directional, achieved by beamforming at both transmitter (Tx) and receiver (Rx) sides. Whereas existing RF human sensing technologies mostly place a subject at zero-degree bearing to ensure sufficient signal quality, our analysis in Section III-A



(a) Main beams. (b) Null point beams.

Fig. 7. Comparing beam patterns for two neighboring arm sites of an angular separation smaller than the width of the main beam.

shows that capturing pulse waveforms at different arm sites with beam scan is possible, but the angle-resolution achieved by the default beamforming scheme is highly inadequate. Therefore, we need to substantially improve the angle-resolution for *h*BP-Fi.

We consider a uniform linear array (ULA) with N Tx antennas spaced by d_{tx} and M Rx antennas spaced by d_{rx}. To scan multiple sites, we program the Tx antennas to successively steer the beam towards different directions in a time-division manner, by emitting RF signals with distinct phase combinations in each time slot. During the k-th time slot, to focus the signals towards a bearing θ, the phase combination for Tx antennas is determined as:

$$\begin{aligned}\vec{\phi}(\theta) &= [\phi_1, \phi_2, \ldots, \phi_N] \\ &= \left[0, \frac{2\pi d_{\text{tx}} \sin\theta}{\lambda}, \ldots, (N-1)\frac{2\pi d_{\text{tx}} \sin\theta}{\lambda}\right], \end{aligned} \tag{6}$$

where λ is the signal wavelength; this endows directionality to the RF sensing. The system FoV is fixed. Limited spatial degrees of freedom can cause beams to partially overlap, reducing variation in the isolated pulse waveforms.

However, it is highly non-trivial to separate the RF reflections from different arm sites: the limited antennas along with a short distance between arm and radar force the reflected signals from main (or even side) beams to get merged together, rendering the boundaries between neighboring sites very vague. For simplicity, we use two main beams to demonstrate the worst case phenomenon of the merged signals for sensing pulse from two neighboring arm sites in Fig. 7(a). Assuming a ULA with 8 Tx-Rx pairs (forming an 8-element virtual array) whose beamwidth is approximately 14°, pulse waveforms from sites #1 and #2 separated by 12° are indistinguishable as the magnitude difference between two main beams is only 1.13 dB [54] (the small power differnce between the yellow and blue curves). Such a phenomenon makes the default radar beamforming scheme insufficient to obtain fine-grained pulse waveforms for deriving accurate BTFs needed by (5). Therefore, we should consider exploring a new method to separate the RF reflections from neighboring arm sites.

To magnify the difference of the RF reflections from two neighboring sites with an angular separation smaller than the width of the main beam, we propose a *differential beamforming with null-steering* scheme to achieve super-resolution beam scan. Null-steering is originally proposed to reject unwanted interference source arriving from a known direction by producing a null point in the response pattern [55], [56]. As shown in Fig. 7(b), in the case of using a ULA with 8 Tx-Rx pairs, two beams separated by 12° yield a remarkable 20 dB amplitude differences between the two main beams, enhancing the separability of pulse signals even under limited angular resolution (bigger difference between the yellow and blue curves).

We present our null-steering via an example of separating RF signals from two configured sites denoted by $\Phi_1(t)$ and $\Phi_2(t)$, respectively. The RF signals received in a time slot can be expressed as [57]:

$$\mathbf{y}_p(t) = \mathbf{v}(\theta_1)\Phi_1(t) + \mathbf{v}(\theta_2)\Phi_2(t), \tag{7}$$

where the steering vector $\mathbf{v}(\theta) = [1, e^{j\alpha(\theta)}, \ldots, e^{j(M-1)\alpha(\theta)}]$ captures the relative phase shifts at each receiver antenna from waves arriving at angle θ, with $\alpha(\theta) = \frac{2\pi d_{\text{rx}} \sin\theta}{\lambda}$.

To isolate the signal from site θ_2, we first construct two sets of weights, $\mathbf{w}_1$ and $\mathbf{w}_2$, forming complementary nulls at $\theta_1 \pm \vartheta$ to suppress the contribution from site θ_1. By assigning these differential weights to the received signals, we obtain:

$$\begin{aligned} s_1(t) &= \mathbf{w}_1^H \mathbf{y}_p(t) \\ &= \left[1, e^{j\alpha(\theta_1)-\vartheta}, \ldots, e^{j(M-1)\alpha(\theta_1)-\vartheta}\right]^\top \mathbf{y}_p(t) \\ s_2(t) &= \mathbf{w}_2^H \mathbf{y}_p(t) \\ &= \left[1, e^{j\alpha(\theta_1)+\vartheta}, \ldots, e^{j(M-1)\alpha(\theta_1)+\vartheta}\right]^\top \mathbf{y}_p(t), \end{aligned} \tag{8}$$

where the weights $\mathbf{w}_1$ and $\mathbf{w}_2$ are designed to create complementary nulls at $\theta_1 \pm \vartheta$, enabling signal separation through their differential response, and $\vartheta = \pi/180$ is a small constant. Associating (7)–(8), we have

$$s_1(t) = a_{11}\Phi_1(t) + a_{12}\Phi_2(t) \quad s_2(t) = a_{21}\Phi_1(t) + a_{22}\Phi_2(t),$$

where $\Delta\theta = 2\pi d_{\text{rx}}(\sin\theta_2 - \sin\theta_1)/\lambda$ and $\{a_{11}, a_{12}, a_{21}, a_{22}\} = \{\sum_{m=0}^{M-1} e^{j\vartheta m}, \sum_{m=0}^{M-1} e^{j(\Delta\theta+\vartheta)m}, \sum_{m=0}^{M-1} e^{-j\vartheta m}, \sum_{m=0}^{M-1} e^{j(\Delta\theta-\vartheta)m}\}$. According to the Euler's formula [58], the complex amplitude a_{11} can be rewritten via

$$\begin{aligned} a_{11} &= e^{j\frac{M-1}{2}\vartheta}\left(e^{-j\frac{M-1}{2}\vartheta} + e^{-j\frac{M-3}{2}\vartheta} \ldots + e^{j\frac{M-3}{2}\vartheta} + e^{j\frac{M-1}{2}\vartheta}\right) \\ &= e^{j\frac{M-1}{2}\vartheta}\left(2\cos\frac{(M-1)\vartheta}{2} + 2\cos\frac{(M-3)\vartheta}{2} + \cdots\right). \end{aligned} \tag{9}$$

Applying the same derivation method, the remaining complex amplitudes can be expressed by the following:

$$\begin{aligned} a_{12} &= e^{-j\frac{M-1}{2}(\Delta\theta+\vartheta)}(2\cos((\Delta\theta+\vartheta)(M-1)/2) \\ &\quad + 2\cos((\Delta\theta+\vartheta)(M-3)/2) + \cdots), \\ a_{21} &= e^{-j\frac{M-1}{2}\vartheta}\left(2\cos\frac{(M-1)\vartheta}{2} + 2\cos\frac{(M-3)\vartheta}{2} + \cdots\right), \\ a_{22} &= e^{-j\frac{M-1}{2}(\Delta\theta-\vartheta)}(2\cos((\Delta\theta+\vartheta)(M-1)/2) \\ &\quad + 2\cos((\Delta\theta-\vartheta)(M-3)/2) + \cdots). \end{aligned} \tag{10}$$

Since a_{11} and a_{21} have the same amplitude but only differ in phase, we can extract $\Phi_2(t)$ by calibrating the phase difference and subtracting the calibrated signals:

$$\begin{aligned}\delta s(t) &= s_2(t)e^{j\frac{M-1}{2}\vartheta} - s_1(t)e^{-j\frac{M-1}{2}\vartheta}\\ &= \Phi_2(t)e^{j\frac{M-1}{2}\Delta\theta}\left(4\sin\frac{(M-1)\Delta\theta}{2}\sin\frac{(M-1)\vartheta}{2}\right.\\ &\quad\left.+4\sin\frac{(M-3)\Delta\theta}{2}\sin\frac{(M-3)\vartheta}{2}+\cdots\right). \quad (11)\end{aligned}$$

where the second term on the right aggregates the trigonometric terms, showing how the small angular offset ϑ amplifies the signal difference. Moreover, $\Phi_1(t)$ can be readily obtained using the same method by replacing θ_2 for θ_1 as the null point in (8). Note that the approach naturally extends to multiple sites by constructing differential pairs for each site combination. This enables the separation of signals from more than two arm locations using the same null-steering principle.

2) Pulse Waveform Extraction: The mmWave radar for *h*BP-Fi adopts FMCW (frequency modulated continuous wave) waveform consisting of a sinusoidal chirp whose frequency increases linearly with time [59], [60]. The pulse-induced skin vibrations lead to minor distance variations that modulate the reflected FMCW signals, affecting their phase over time. On the receiving side, a mixer multiples the FMCW reflections at time t with the signal transmitted at time t to downcover it. The downconverted signals are then processed by range FFT to produce a range profile composed of multiple range bins, each corresponding to a different reflection distance [61]. We identify the bin corresponding to the subject's arm (with the strongest reflection) as the tracking target. This is followed by differential beamforming with null-steering to further separate the data from different arm sites. By aligning this bin across successive chirps, we form a slow-time sequence that captures micro-movements due to pulsation. The complex-valued signal from this bin (at frequency f_k) yields a phase profile $\phi^j(t, f_k)$ of each antenna j. Specifically, $\phi^j(t, f_k)$ is given by $2\pi\frac{d(t)}{\lambda}$, where λ is the wavelength of the transmitted signal, and $d(t)$ is the round trip distance between the reflector and the device. As $d(t)$ changes with arm pulses, the phase waveform encodes this variation.

We measured pulses from two seated subjects wearing a winter jacket and a T-shirt, following the method above. Fig. 8(a)–(d) show example pulse waveforms from two arm sites separated by 12° (different colors represent different sites). While the waveforms from the two sites are clearly distinct in all cases, differences across subjects and scenarios introduce additional ambiguities. Based on these waveforms, we measured BP by searching for the optimal parameter set (as described in Section III-A). As a pilot study, we compared 50 pairs of measured BPs with cuff-based ground truth (Fig. 8(e) and (f)). The results show strong correlation, confirming the feasibility of the BTF model for BP inference. However, differences in accuracy across subjects and scenarios highlight the need for cross-subject/scenario methods, favoring a data-driven approach.

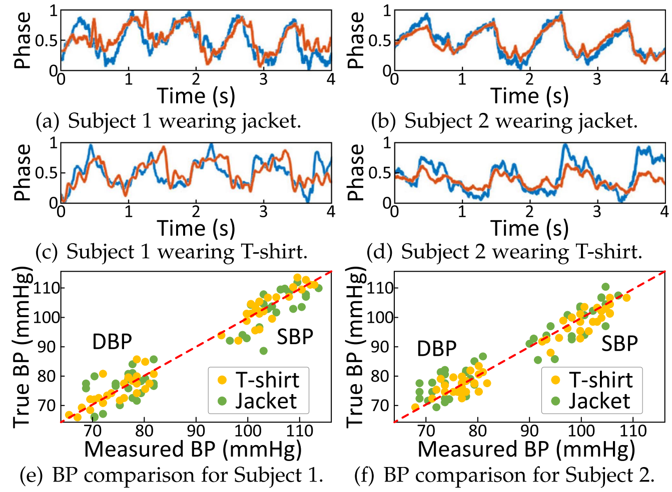


Fig. 8. Examples of pulse waveforms obtained from two subjects under two scenarios (a-d), and comparisons between BP inferred via BTF and true BP (e-f).

3) Interference Detection and Removal: While RF signal's sensitivity to distance variations enables effective extraction of pulse signals; however, it also causes strong RF responses from other body movements (e.g., arm waving), which can completely overshadow subtle pulses. Thus, we advise users to minimize physical activity during system operation in our early version [62]. However, in practice, even slight posture shifts from breathing/heartbeat cause subtle interference that overlaps with pulse signals. Therefore, it is necessary to identify and mitigate these low-magnitude interference components. Although waveform feature analysis and template matching are effective for detecting motion artifacts in conventional heartbeat signals like PPG and ECG [63], they are impractical for radar-captured pulse signals along the arm due to the variability in waveforms across different arm positions and differing measurement setups. To address this, we propose a two-step approach: first, we estimate the heartbeat interval and segment cardiac cycles; second, we check the signal periodicity within each cycle to detect interference:

Step 1: we employ the widely validated autocorrelation-based method [64] to estimate the heartbeat interval. Within a cardiac cycle, the arm pulse waveform typically features a prominent peak (as shown in Fig. 3). Based on this, within each processing window, we locate the highest peak, then iteratively search for preceding and subsequent prominent peaks, guided by the estimated heartbeat interval, until the total number of identified peaks matches the expected number of heartbeat cycles. These peaks serve as delimiters, allowing us to segment the pulse signal into individual heartbeat episodes.

Step 2: To detect interference, we compare adjacent heartbeat episodes based on the similarity of their waveform, where clean signals exhibit high similarity (indicating periodicity nature), while noisy signals show reduced similarity. Specifically, Pearson correlation coefficient is used to quantify this, with signals considered similar if their coefficient exceeds 0.8 (a well-established threshold supported by prior research [65], [66], [67]); otherwise, they are flagged as potentially affected

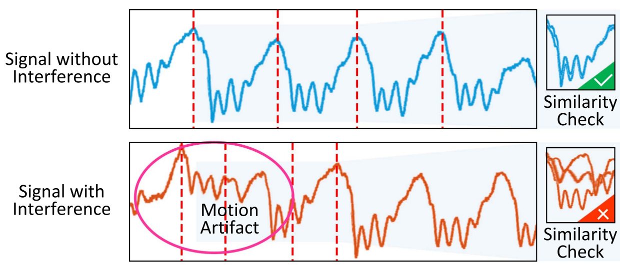


Fig. 9. Interference detection based on periodicity.

by interference. This concise and efficient method allows for rapid identification of disturbed segments. Fig. 9 exemplifies the effective differentiation between signals with and without interference, wherein dashed red lines delineate the boundaries of heartbeat intervals based on the identified prominent peaks.

C. BP Inference

Existing wireless-based BP inference networks lack physiological basis, rendering their inference opaque and difficult to interpret. To address this, we design an interpretable DL network that improves structural transparency and output intelligibility. To further enhance user experience and system robustness for unseen users and scenarios, we extend this interpretable network into a CycleGAN-based framework, where competitive mechanisms enhance both interpretability and network effectiveness.

1) Interpretable Neural Pipeline: The combination of mathematical model and DL have been found effective in various tasks [68], [69], and are typically achieved by cascading them: first build a rough mathematical model to initialize the results then use data-driven DL to refine them, or first develop data-driven DL to estimate a rough approximate solution then cascading a physical model-driven network to refine this into an exact solution [70]. However, both solutions separate the DL training from mathematical model resolving, entailing manually tedious parameter adjustments yet obtaining only shallow feature priors [70]. To overcome these issues, we integrate a mathematical BTF model with DL by designing the network architecture based on the BTF solving process and incorporating BTF into the loss function, thereby endowing both the network structure and the optimization space with interpretability.

We build *h*BP-Fi's BP estimator via deeply-recursive convolutional network (DRCN) based on the solving process of the BTF, then incorporated long short-term memory (LSTM) modules to mitigate temporal and thermal variations, thereby achieving accurate estimation of BP waveform ρ. Specifically, BTFs are first measured from the time series of varying pulse waveforms via $\Gamma_{\mathrm{i}\to\mathrm{o}} = P_{\mathrm{o}}(t)P_{\mathrm{i}}^{-1}(t)$. As shown in Fig. 10, the BP estimator takes the measured BTFs as input, then it splits the optimization of (5) into sub-problems based on alternating direction method of multipliers (ADMM) [71] as follows:

$$\Theta^{t+1} = \Theta^t - \eta \frac{\partial}{\partial \Theta^t} J(\Theta) + \frac{\gamma}{2} \left\|\lambda^t + A\Theta^t + BZ^t - C\right\|_2^2$$
$$Z^{t+1} = \mathbf{DRCN}(\Theta^{t+1}, \lambda^t)$$

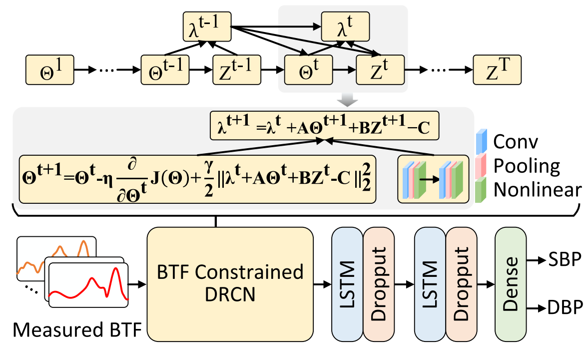


Fig. 10. The neural pipeline of *h*BP-Fi's BP estimator with the BTF-based DRCN structure.

$$\lambda^{t+1} = \lambda^t + A\Theta^{t+1} + BZ^{t+1} - C, \tag{12}$$

which is subject to $A\Theta + BZ = C$, where A, B, and C are ADMM matrices whose dimensions depend on the shapes of Θ and Z [71]. In addition, λ is the Lagrange multiplier and γ is the penalty parameter. Each iteration is unfolded into one sub-network in recursion, which forms multi-cascaded sub-networks sharing the same network parameters. The BTF-based DRCN integrates the BTF model with DL by automating iterative optimization and variable splitting, thereby improving the transparency and interpretability of the architecture. Through this iterative process, the network estimates the BP waveform ρ, whose peak and trough theoretically correspond to SBP and DBP. We note that residual reflections might introduce minor amplitude and phase perturbations in the BTF, but the main trend is preserved, as the iterative optimization effectively mitigates their impact.

Because these extremum values are highly sensitive to noise and physiological variability, the estimated waveform is further refined by LSTM modules to capture stable temporal patterns. A dual-branch regression head then uses these refined features—via a shared encoder and two task-specific heads—to produce robust SBP and DBP predictions. The loss function for estimation commonly adopts the mean square error (MSE) to minimize the difference between the network output and the ground truth values, which could be susceptible to interference from outlier BP values or measurement noise, potentially biasing the model optimization away from the true underlying distribution. As *h*BP-Fi is concretely based on hemodynamics, we additionally take into account the error of the BTF optimization to establish the BTF-constrained loss function:

$$\mathfrak{L}(E_\rho) = \|J(\Theta)\|_2^2 + \|A\Theta + BZ - C\|_F^2 + \|\rho - \tilde{\rho}\|_2^2, \tag{13}$$

where $\|\cdot\|_F$ represents the Frobenius norm, $\rho = \{\rho_{\mathrm{s}}, \rho_{\mathrm{d}}\}$ and $\tilde{\rho} = \{\tilde{\rho}_{\mathrm{s}}, \tilde{\rho}_{\mathrm{d}}\}$ denote the ground truth values and estimated values of SBP and DBP, respectively. This loss function takes the BTF-constrained parameter optimization as a regularization term, rendering the optimization space interpretable.

2) Cross-Domain Training Via CycleGAN: In order to have the BP estimator calibration-free, the training procedure typically requires a high volume of training data, labeled by true SBP and DBP values, under a wide range of measurement domains, or

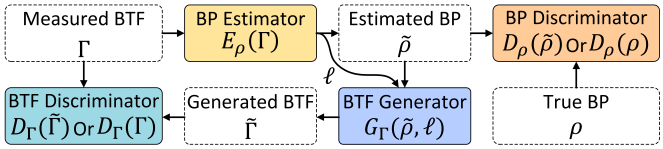


Fig. 11. DL training framework based on CycleGAN.

otherwise relying on transfer learning technology [7]. However, these methods are not feasible for training *h*BP-Fi, due to the limited data availability and the strong assumption of transfer learning on data similarity between source and target domains. To this end, we leverage the power of CycleGAN [72] to enforce the cross-domain capability to *h*BP-Fi without the need for the aforementioned conditions or assumptions. CycleGAN was originally proposed to achieve unpaired image-to-image translation, which typically consists of two generators and two discriminators. The rationale is to respectively learn a translation and an inverse translation between two distributions, with the goal of reducing the difference between translated data and true data, hence enforcing the generators to produce plausible results without paired data from the two domains.

Our CycleGAN training architecture is shown in Fig. 11, where the BP estimator E_ρ takes the BTFs Γ measured from varying pulse waveforms along the arm as input and estimates BP values $\tilde{\rho}$ as output on one direction via $\tilde{\rho} = E_\rho(\Gamma)$, and the estimated BP $\tilde{\rho}$ and other BTF model parameters ℓ are given as input to the BTF generator G_Γ to produce BTFs $\tilde{\Gamma}$ on the reversed direction via $\tilde{\Gamma} = G_\Gamma(\tilde{\rho}, \ell)$. Following the BP estimator and BTF generator are two discriminators D_ρ and D_Γ that complete the cyclic training process. The unpaired training forces the proposed architecture to learn stronger scenario-independent and subject-independent mapping between BTFs and BPs, making *h*BP-Fi a robust BP estimator across unseen domains with limited training data.

Specifically, the BTF generator aims to generate domain-independent BTFs based on the estimated BP and BTF parameters solved by BP estimator. We adopt attention U-Net [73] as the BTF generator architecture recently gaining success in physiological signal reconstruction [74]. The encoder of the BTF generator consists of 5 blocks using 64, 128, 256, 512, and 512 filters, respectively, and the kernel size is 1×16 and the stride is 2. Except for the first block, each block incorporates self-gated soft attention with layer normalization and *ReLU* activation after the convolutional layer to emphasize significant features conveyed through the skip connections. The decoder adopts a similar architecture as the encoder, but with the filter quantities reversed and de-convolution layers using *ReLU* activation applied.

BP and BTF discriminators are built to determine whether the input comes from estimated (resp. generated) results or true data gathered in the real world. The BP discriminator takes estimated BP and true BP as inputs, while the BTF discriminator takes generated BTF and measured BTF as inputs. Both discriminators utilize a common architecture consisting of four convolution layers with varying numbers of filters (64, 128, 256, and 512) for feature extraction, followed by a dense layer with *softmax* activation to output the probability of the input data belonging to the actual BP/BTF dataset.

The overall loss of our CycleGAN architecture is a combination of the adversarial loss and the cyclic consistency loss. Specifically, we denote the ground truth of BP by $\rho = \{\rho_\mathrm{s}, \rho_\mathrm{d}\}$. The adversarial loss of the BP estimator E_ρ and reliability predictor P_ρ is defined as follows:

$$\mathcal{L}_\mathrm{adv}(E_\rho, D_\rho) = \mathbb{E}[\log(D_\rho(\rho))] + \mathbb{E}[\log(1 - D_\rho(\tilde{\rho}))]. \quad (14)$$

Similarly, the adversarial loss of the BTF generator G_Γ and BTF discriminator D_Γ is defined as follows:

$$\mathcal{L}_\mathrm{adv}(G_\Gamma, D_\Gamma) = \mathbb{E}[\log(D_\Gamma(\Gamma))] + \mathbb{E}[\log(1 - D_\Gamma(\tilde{\Gamma}))]. \quad (15)$$

The cyclic consistency loss quantifies the BP estimation loss and BTF reconstruction loss:

$$\mathcal{L}_\mathrm{cyc}(E_\rho, G_\Gamma) = \mathfrak{L}(E_\rho(\tilde{\Gamma})) + \mathbb{E}\left[\left\|\tilde{\Gamma} - \Gamma\right\|_2\right]. \quad (16)$$

Finally, the overall loss can be represented as

$$\boldsymbol{\mathcal{L}} = \mathcal{L}_\mathrm{adv}(E_\rho, D_\rho) + \mathcal{L}_\mathrm{adv}(G_\Gamma, D_\Gamma) + \beta\mathcal{L}_\mathrm{cyc}(E_\rho, G_\Gamma). \quad (17)$$

IV. System Evaluation

This section presents the implementation and experimental setup of *h*BP-Fi, followed by a comprehensive evaluation under various parameter settings.

A. Prototyping and Experiment Setup

1) Hardware and Software: *h*BP-Fi adopts an mmWave radar as the sensing front end for simultaneously monitoring varying pulse waveforms along the arm, and a laptop is used as the data processing back end. The adopted radar is a commercially available TI AWR1843BOOST mmWave radar [23]. It operates at 77 GHz, with 3 Tx antennas and 4 Rx antennas, and a bandwidth of up to 4 GHz. A TI DCA 1000EVM data capture adapter [75] is connected to the radar to acquire real-time data and stream them to the laptop. During data collection, the frame sampling rate is set to 200 frames/second. *h*BP-Fi's data processing pipeline is implemented in MATLAB R2019b and Python 3.8. The CycleGAN pipeline is built upon TensorFlow 2.2. We first processed the captured FMCW reflections as described in Section III-B2 to obtain pulse waveforms, then apply a sliding window of 512 samples shifted by 64 samples to infer BP.

2) Data Collection: This study is approved by our institute's ethics committee. We recruit 35 participants with diverse physiological characteristics, covering a wide range of ages (20–54), sexes (22 males and 13 females), and blood pressure levels (SBP: 90–130 mmHg; DBP: 60–80 mmHg). This diversity ensures broad coverage of the general population.

The experiment setup is shown in Fig. 12, all measurements follow the standard validation protocol for BP monitoring [25] under expert supervision. Subjects sit with their backs supported, legs uncrossed, and—critically—their arms extended horizontally and kept orthogonal to the torso. This posture ensures the radar directly faces the arm while preventing respiratory or cardiac motions from contaminating the radar reflections. Each session consists of 4–8 sets of pulse waveforms, system outputs,

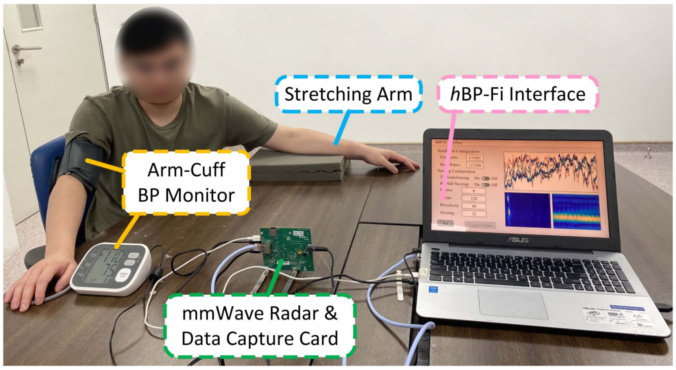


Fig. 12. Experiment setup with all devices exhibited.

and reference BP values. Each recording lasts at least 60 s, repeated at 1–2 min intervals, with at least 5 min of rest between rounds. BP values are validated through averaging and consistency checks. To ensure high signal fidelity, we combine posture control, minimized body movement, and optimized radar placement. The mmWave radar is positioned directly in front of the subject with its FoV fully covering the arm while shadowing the torso and other body parts, thereby suppressing physiological interference. It continuously scans pulse waveforms along the arm, while residual motion artifacts (e.g., subtle breathing) are detected and removed using interference-filtering algorithms.

Participants complete data collection across 12 typical daily scenarios combining four clothing types (T-shirts, sweaters, hoodies, winter jackets) and three tabletop materials (wood, laminate, glass). Additional measurements are taken under controlled indoor temperatures ($23 - 28°$C) between 9:00 AM and 4:00 PM. Since the relative angle and distance[2] between the arm and the radar device may vary during actual use, we also evaluate *h*BP-Fi under different radar placements, while the elevation angle is fixed to ensure full coverage of the arm.

Ground truth BP is obtained using an FDA-approved arm-cuff BP monitor Omron 7127 [76], with the cuff held at heart level. This device is used only for labeling and is not part of *h*BP-Fi during deployment. Overall, we collect approximately 252,000 heartbeat cycles from all participants. Moreover, given the relatively concentrated distribution of ground truth BP data, we augmented the collected dataset by replicating it 10 times with different random noises. This procedure effectively increases data diversity and helps reduce overfitting risk. To prevent any risk of data leakage, the dataset is strictly partitioned such that no data from the same subject or scenario appears simultaneously in both training and testing sets.

3) Evaluation Methodology: To evaluate the performance of *h*BP-Fi, we use two widely adopted metrics, i.e., mean error $\mathrm{ME} = \sum_{i=1}^{n}(\tilde{\rho}_i - \rho_i)/n$ and standard deviation of mean error $\mathrm{STD} = \sqrt{\sum_{i=1}^{n}(\tilde{\rho}_i - \rho_i - ME)^2/n}$, where $\tilde{\rho}$ and ρ respectively denote the estimated and ground truth (true) BP values.

[2] The distance is measured from the radar antenna's center to the robotic arm's measurement plane, while the angle is formed between the arm's axis and the radar's plane. During setup, the radar is positioned first, then the antenna center marks the arm's placement for precise alignment.

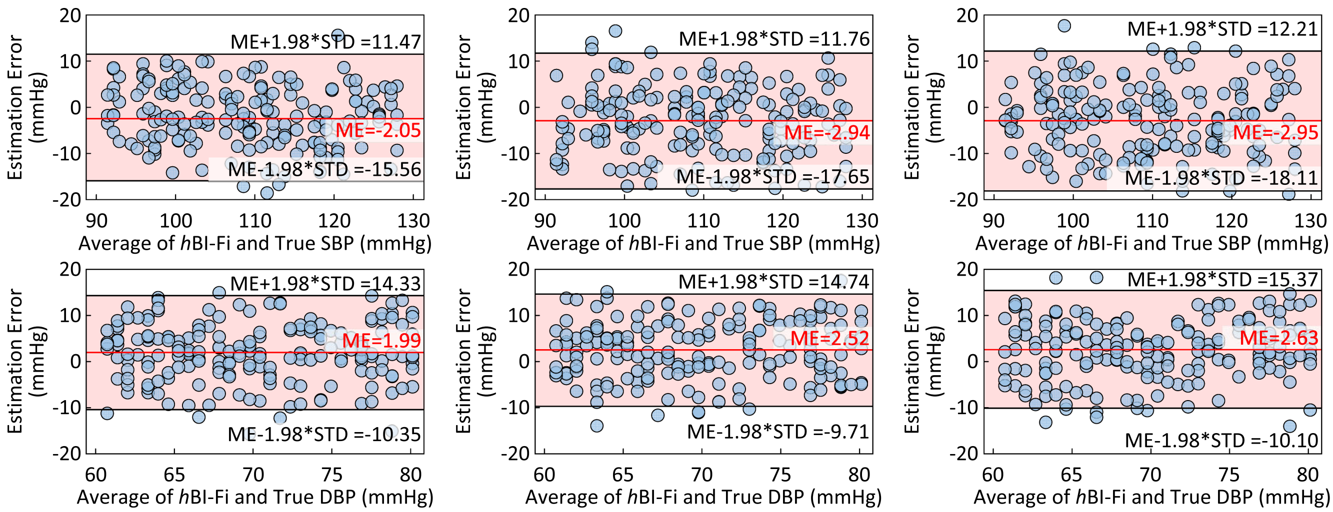
B. System-Level Evaluation

1) Overall Performance: We first study *h*BP-Fi's ability in estimating BP leveraging RF-sensed pulse waveforms. Specifically, *h*BP-Fi is designed to be adaptable to unseen subjects/scenarios. To validate these essential claims, we conduct three experiments involving different combinations of subjects and scenarios. These three experiments collectively highlight the strict subject-scenario isolation strategy we adopted to avoid data leakage and overfitting, in contrast to prior studies that often evaluate on overlapping data.

First, we conduct a rigorous leave-one-subject-out experiment, where data from one subject is used for testing while data from the remaining subjects is used for training. This ensures that the training and testing sets are completely disjoint in terms of individuals. Results from all combinations are illustrated in Fig. 13(a), which shows the Bland-Altman plots for the estimated SBP and DBP. More than 95% points are narrowly distributed within the limits of agreement, i.e., ME $\pm 1.98 \times$ STD, suggesting that the estimated BP can be an alternative to the true BP. Overall, *h*BP-Fi achieves the ME of -2.05 mmHg and STD of 6.83 mmHg for estimating SBP and ME of 1.99 mmHg and STD of 6.30 mmHg for estimating DBP. Such decent performance indicates that *h*BP-Fi can be well adapted to unseen subjects.

Due to the strong penetration capability of mmWave signals through clothing and the distinct separation between static desktop reflections and dynamic pulse signatures, the acquired signals should exhibit high quality with negligible variations across different scenarios. If this holds, models trained on data from diverse scenarios should generalize well to unseen settings. To evaluate generalization across environments, we conduct leave-one-scenario-out experiments. Here, data from one scenario is reserved for testing, while data from all other scenarios is used for training. This setup guarantees that the testing environment is never seen during training, thereby isolating the impact of contextual differences. The results are shown in Fig. 13(b). We may observe that the points have a narrow distribution within ME $\pm 1.98 \times$ STD margin, demonstrating that *h*BP-Fi obtains comparable BP results with the arm-cuff BP monitor. Specifically, *h*BP-Fi achieves -2.94 mmHg ME and 7.43 mmHg STD for estimating SBP and ME of 2.51 mmHg and STD of 6.24 mmHg for estimating DBP. The low MEs and STDs confirm *h*BP-Fi's robustness to scenario variations.

Finally, we are interested in how *h*BP-Fi would perform with a new subject in a new scenario. Therefore, we re-group the dataset and use data from one subject in one scenario for testing, and data from the remaining subjects in other scenarios for training; this leave-one-*domain*-out experiment ensures that the training and testing are mutually exclusive and that *h*BP-Fi is evaluated on subjects and scenarios unseen during training. Fig. 13(c) shows the results with more than 95% of the points in the limits of agreement, validating that the estimated BP and true BP can be used interchangeably. Specifically, *h*BP-Fi achieves the ME of -2.95 mmHg and STD of 7.66 mmHg for estimating SBP and ME of 2.63 mmHg and STD of 6.50 mmHg for estimating DBP, demonstrating that *h*BP-Fi successfully handles domain



(a) Leave-one-subject-out. (b) Leave-one-scenario-out. (c) Leave-one-domain-out.

Fig. 13. Overall performance in terms of its generalizability to unseen subjects and scenarios.

TABLE II
COMPARING *h*BP-FI WITH AAMI STANDARD

		ME (mmHg)	STD (mmHg)
*h*BP-Fi	SBP	-2.95	7.66
	DBP	2.63	6.50
AAMI	SBP and DBP	$\leqslant 5$	$\leqslant 8$

TABLE III
COMPARING *h*BP-FI WITH BHS STANDARD

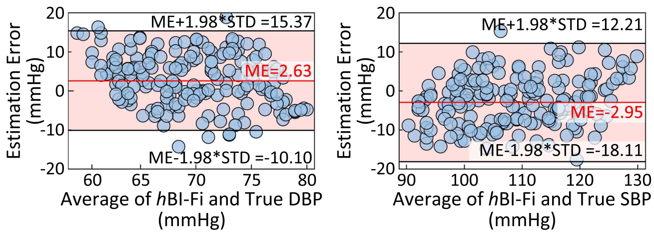
		Cumulative Error Percentage		
		$\leqslant$5 mmHg	$\leqslant$10 mmHg	$\leqslant$15 mmHg
*h*BP-Fi	SBP	61.50%	86.63%	97.65%
	DBP	62.57%	87.82%	98.93%
BHS	Grade A	60%	85%	95%
	Grade B	50%	75%	90%
	Grade C	40%	65%	85%

variations and enables adaptation to unseen subjects under unseen scenarios. By analyzing the results of the above three experiments, we can find that *h*BP-Fi exhibits negative mean errors in estimating SBP (indicating a tendency toward underestimation), while demonstrating positive mean errors for DBP (suggesting a tendency toward overestimation). Upon careful examination, we attribute these biases primarily to distributional mismatches between training and test data, a limitation that presents an important direction for future enhancement. In summary, the promising results confirm that *h*BP-Fi is capable of being an alternative solution for accurate BP monitoring and of generalizing to unseen subjects/scenarios. Encouraged by the performance in these controlled settings, the final model for deployment is retrained using the full dataset with the selected architecture and hyperparameters.

To further validate *h*BP-Fi, we compare the results of leave-one-domain-out experiment with the acceptable range regulated by the Association for the Advancement of Medical Instruments (AAMI) and the requirement for the Britain Hypertension Society (BHS) standard [77] in Tables II and III, respectively. One may readily observe that *h*BP-Fi satisfies the recommended error boundary defined by the AAMI and achieves BHS Grade A for estimating DBP and SBP. While the AAMI recommends a test cohort of 85 subjects [25], we acknowledge that our current study includes 35 subjects due to practical limitations. Nevertheless, the leave-one-domain-out protocol rigorously evaluates generalization to unseen domains, offering a conservative and meaningful proxy to large-cohort testing. It is worth noting that a recent proposal mmBP [17] claims to achieve slightly lower estimation errors. However, mmBP relies on a black-box DL model that directly maps single-site pulse measurements to BP values, without physiological interpretability. Furthermore, its evaluation may suffer from data leakage by testing on subjects and scenarios seen during training. In contrast, *h*BP-Fi is grounded in physiological insights and implemented via an interpretable DL model, enabling robust generalization to new users and settings.

2) Impact of Environment Factors: We are interested in how *h*BP-Fi would perform in diverse real-world environments. Therefore, we group the dataset into 12 environment groups based on temperature ($23-28^{\circ}$C, binned at 2°C intervals) and time of day (9:00 AM–4:00 PM, binned in 2-hour intervals). Adopting a leave-one-environment-out cross-validation strategy, we reserve data from one environment for testing while training on the remaining eleven environments, ensuring that the training and testing are mutually exclusive and that *h*BP-Fi is evaluated in an untrained environment. Fig. 14 shows the results with more than 95% of the points in the limits of agreement. Specifically, *h*BP-Fi achieves the ME of -2.95 mmHg and STD of 7.66 mmHg for estimating SBP and ME of 2.63 mmHg and STD of 6.50 mmHg for estimating DBP. The promising results confirm that *h*BP-Fi is capable of being an alternative solution for accurate BP monitoring and can successfully handle environmental factor variations.

3) Impact of Radar Placement: In real-world environments, the distance and angle between the subject's arm and the radar may be different. Given that arm-to-radar distance and angle



(a) DBP. (b) SBP

Fig. 14. System performance in leave-one-environment-out experiment.

TABLE IV
PERFORMANCE WITH DIFFERENT RADAR PLACEMENTS

	ζ \ d	30 cm	50 cm	70 cm	90 cm
SBP	0°	-2.95±7.66	-2.94±7.72	-3.25±8.21	-4.37±8.12
	15°	-2.90±7.64	-2.93±7.67	-3.05±8.36	-4.36± 8.61
	30°	-2.92±7.61	-2.97±7.91	-3.38±8.11	-4.36±7.79
	45°	-2.96±7.82	-2.95±7.50	-3.00±8.20	-4.38±8.77
DBP	0°	2.63±6.71	2.93±6.82	3.68±7.47	4.33±7.22
	15°	2.67±6.20	2.94±6.51	3.69±7.63	4.34±7.30
	30°	2.69±6.85	2.99±6.53	3.68±7.78	4.29±7.72
	45°	2.63±6.44	2.96±7.72	3.69±7.28	4.36±7.87

vary in practice, we evaluate these parameters to characterize the operational range for reliable BP monitoring, testing our hypothesis that greater distances and angles would impair performance through SNR reduction. Specifically, we ask all subjects to collect data with distance $d = \{30\text{cm}, 50\text{cm}, 70\text{cm}, 90\text{cm}\}$ and angle $\zeta = \{0°, 15°, 30°, 45°\}$. Table IV summarizes the ME and STD quantified by the above factors. Overall, the performance of *h*BP-Fi is stable as the angle changes. Besides, we notice that the SBP and DBP are negatively affected by the increasing sensing distance. Encouragingly, the DBP remains within the acceptable range regulated by the FDA protocol when the distance is within 70 cm, while the SBP falls within the acceptable range when the distance is within 50 cm. We specifically gathered feedback about measurement distances. Participants generally preferred closer positioning to facilitate alignment of the arm and the device, with two preferring 30 cm while others accepted 50 cm; therefore, *h*BP-Fi has confirmed robustness in real-world adoption.

C. Micro-Benchmark Evaluation and Ablation Study

1) Ablation on Super-Resolution Beam Scan (SRBC): SRBC is essential for capturing fine-grained BTFs and enabling accurate BP estimation. To validate its contribution, we perform an ablation experiment by replacing SRBC with a traditional beamforming scheme [61]. In this scheme, data is collected independently at each beam direction, with beamforming applied only in post-processing to ensure fair comparison under identical hardware constraints. This isolates the contribution of SRBC to overall system performance. Pulse waves from specific angle bins [61] are treated as corresponding arm-site pulses. Fig. 15 compares SBP and DBP estimation errors between the two schemes, where points represent ME and error bars indicate

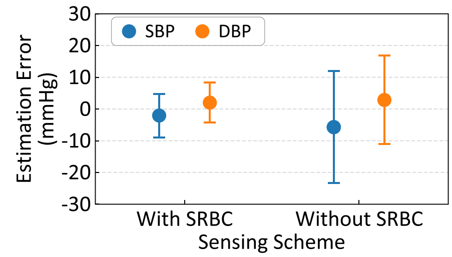
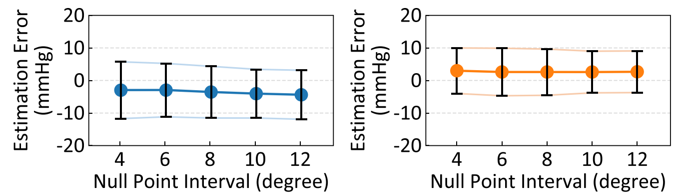


Fig. 15. Performance with different sensing schemes.



(a) Estimation error for SBP (b) Estimation error DBP

Fig. 16. Impact of null point intervals.

STD. Without SRBC, the errors are -5.72 ± 17.65 mmHg and 2.89 ± 14.01 mmHg for SBP and DBP, respectively, which are reduced by SRBC to -2.95 ± 7.65 mmHg and 2.63 ± 6.50 mmHg. These results demonstrate that SRBC effectively captures fine-grained BTF features, improving the robustness and accuracy of BP estimation.

2) Impact of Null Point Distribution: As described in Section III-B1, we set multiple null points evenly spaced along the arm, i.e., spaced at an identical angle, to separate pulse waveforms from specific arm sites. More null points may result in too little variation between pulse waveforms, making it difficult to characterize hemodynamics. Fewer null points, on the other hand, would result in fewer pulse waveforms being captured along the arm, which could hinder BTF optimization. To study the impact of null point distribution on *h*BP-Fi's performance, we particularly collect data at intervals of 4°, 6°, 8°, 10°, and 12° between null points, all finer than the width of the main beam of 14°. Fig. 16 shows both MEs and STDs for SBP and DBP estimates; one may observe that i) MEs do not vary significantly from intervals 4° to 6°, while STD for SBP slightly decreases while the STD for DBP slightly increases, ii) MEs and STDs decrease noticeably after the interval reaches 8° and 10°, and iii) MEs and STDs become relatively saturated till the interval of 12°. Using the interval of 10° or 12° achieves good performance, but 12° is chosen as larger intervals can save computational resources.

3) Ablation Study on Interference Detection and Removal: Although participants made every effort to remain physically still during measurements to avoid intense motion artifacts, subtle physiological movements, such as those induced by respiration or cardiac activity, are unavoidable. Despite these strict controls, approximately 7% of measurements were still affected by minor artifacts (e.g., tremors), as detected by our interference detection algorithm. To assess the necessity and effectiveness of our interference detection and removal algorithm, we perform an ablation study by evaluating *h*BP-Fi under three cases: i)

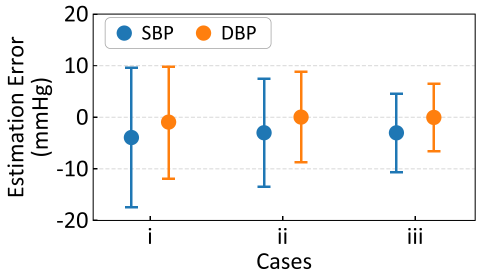


Fig. 17. Impact of interference detection.

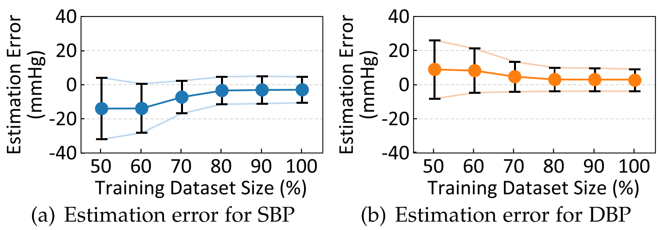


Fig. 18. Impact of training dataset sizes.

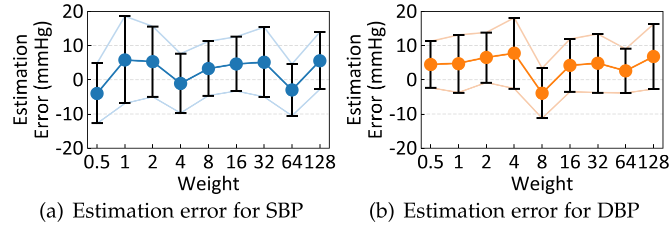


Fig. 19. Impact of weight values.

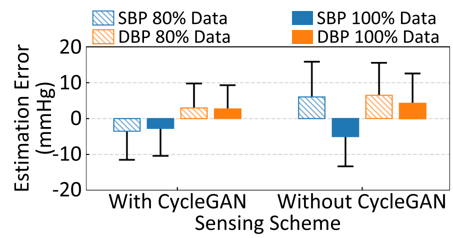


Fig. 20. Impact of CycleGAN-based architecture.

testing only instances with interference, ii) testing all collected instances, including those with interference, and iii) testing collected instances excluding interfered instances. The results presented in Fig. 17 indicate that instances with interference (case i) substantially elevate BP estimation errors. A comparison between case ii and case iii reveals that retaining interfered instances in the testing data results in a slight decrement in overall performance. However, since the proportion of interfered instances is low, the overall performance remains very close to the acceptable medical boundaries, with SBP error of -3.02 ± 10.41 mmHg and DBP error of 2.37 ± 8.74 mmHg. Nonetheless, this ablation highlights the necessity of interference handling: as any high-error readings could lead to unwarranted stress or incorrect medication administration for users.

4) Impact of Training Dataset Size: Theoretically, a larger training data set would result in a better DL model. To study how training dataset size affects model performance, we conduct a leave-one-domain-out experiment while ensuring the diversity of training data remains constant. Specifically, all subjects and all scenarios are included in the training set, and only the amount of data per subject per scenario is varied (50%, 60%, 70%, 80%, 90%, and 100%). Fig. 18 shows MEs and STDs for SBP and DBP estimates with varying training data sizes. One can observe that the MEs for SBP and DBP decrease by 8.98 mmHg and 3.59 mmHg, respectively, as the training dataset size increases from 50% to 80%. Then MEs come to saturation as the training dataset size increases to 90% and 100%. As STDs follow exactly the same trend, we refrain from making detailed comments. These results indicate that our currently used training dataset is sufficient, and adding more training data only improves the BP estimation performance by a negligible margin.

5) Impact of Weight of Cyclic Consistency Loss: The weight of cyclic consistency loss involved in the total loss of the CycleGAN framework, i.e., β in (17), is a crucial parameter to be tuned. In theory, larger β prevents the BP estimator from excessive hallucinations and mode collapse (both of which will cause unnecessary loss of information) but could cause unwanted artifacts [78]. To determine the optimal weight, we compare both MEs and STDs for SBP and DBP with varying β values in Fig. 19. It can be observed that MEs and STDs fluctuate as β increases but remain (for both SBP and DBP) between -5 $\pm$ 10 and 5 $\pm$ 10 when $\beta = 0.5$, 8, and 64, which is within the range required by AAMI. We carefully study the performance of these three conditions and find that $\beta = 64$ leads to the lowest estimation error, thus 64 is used in our DL pipeline.

As described in Section III-C2, we develop the CycleGAN-based architecture to achieve a combination of generalization and training-effective, i.e., using a reasonable amount of training data to build an effective model that can adapt well to unknown domains. The experiments presented in Sections IV-B1, IV-B3, and IV-C4 evidently confirm the effectiveness of *h*BP-Fi for cross-domain BP estimation, while confirming the amount of training data needed. Herein, we conduct an ablation study to verify how much the CycleGAN architecture contributes to achieving the excellent performance of *h*BP-Fi. Fig. 20 compares the MEs and STDs for SBP and DBP, with/without using the CycleGAN architecture. It can be observed that i) using the CycleGAN architecture reduces MEs in all cases, and ii) even without using the CycleGAN architecture, *h*BP-Fi still achieves results around the AAMI range, which we believe owes to the DRCN's accurate solution of the BTF model. Whereas when using the CycleGAN-based architecture, *h*BP-Fi achieves decent results using a smaller amount of training data. Such results validate that the proposed CycleGAN architecture effectively handles domain dependence, making it adaptive to new domains with a reasonable amount of training data.

6) Interpretability in DL: Despite the clarification of DL model's interpretability in Sec. III-C1, we further demonstrate the interpretability in DL by showcasing how changes in the inputted BTFs are associated with variations in the estimated

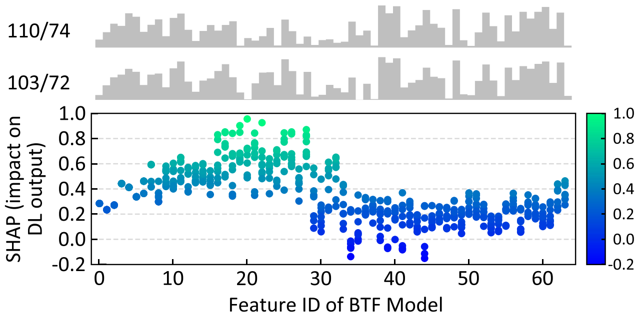


Fig. 21. The SHAP plot of features from BTF.

BP values, providing insights into the BP-estimating process. To achieve this, we employ *shapley additive explanations* (SHAP) [79] to present visual interpretability of the impact of different features on BP measurement. Specifically, each BTF input is divided into 64 groups, where the mean of each group serves as a feature, and SHAP values are computed using SHAP gradient explainer. The top panel of Fig. 21 illustrates BTF features for two instances of SBP/DBP values from the same user: 110/74 and 103/72, while the bottom panel shows the corresponding SHAP values, where higher SHAP values indicate more significant contributions to BP variations. It can be observed that features #10 to #30 exhibit significant variations accompanied by higher SHAP values, while the remaining features show minimal changes with lower SHAP values. This direct association between the changes in BTF features and BP variations establishes a clear linkage, addressing the challenge of uncorrelated BP changes with BTF variations in the black-box DL setting. By assessing the consistency of BTF features with medical literature, such as prospective clinical trials and physiological mechanisms, users and clinicians can validate the BP results.

V. Limitations and Future Work

As the first contactless BP monitoring system built upon interpretable, deeply analyzed hemodynamics, *h*BP-Fi represents an important step toward timely and regular health monitoring. Nevertheless, several limitations remain and motivate future research. *Clinical Validation and Participant Diversity:h*BP-Fi currently recruits only healthy participants due to IRB constraints, limiting the BP range to normotensive levels (SBP 90–130 mmHg, DBP 60–80 mmHg) and restricting clinical generalizability. The reported high accuracy reflects these controlled conditions, and real-world factors such as motion, posture variations, and environmental interference may increase error rates. While a certified oscillometric monitor serves as the reference device, future large-scale studies across diverse ages, genders, and clinical conditions are underway to validate performance in broader populations.

*Sensing Robustness: h*BP-Fi currently experiences performance degradation when the sensing distance approaches 80 cm due to single-radar attenuation. The system also assumes minimal body motion and does not account for vertical angular offsets, such as pitch variations. Moreover, reflections from other sites may introduce minor interference in the extracted BTF signals. These factors limit robustness in unconstrained environments. Future work will explore multi-radar fusion to extend room-scale sensing, DL–based denoising and spatiotemporal modeling to tolerate body motion, and optimized beamforming to handle diverse arm–radar geometries.

BTF Reliability and Extension While the results demonstrate the feasibility of extracting BTF for contactless BP estimation, further evidence is needed to validate its reliability across broader physiological conditions. Future studies will examine additional complementary hemodynamic features to enhance interpretability and ensure consistent performance across individuals.

Model Optimization and Trustworthiness Our current implementation selects hyperparameters empirically based on preliminary experiments. To improve reproducibility and generalization, future work will investigate systematic optimization methods such as grid search, Bayesian optimization, and population-based training. Moreover, to enhance the trustworthiness of BP predictions, *h*BP-Fi will incorporate auxiliary metrics such as confidence levels, fuzzy scores, or credibility indices, providing clinicians and users with interpretable indicators of prediction reliability.

On-Device Deployment Because mmWave radars are already embedded in millions of smartphones, *h*BP-Fi is compatible with existing mobile hardware, enabling seamless integration for continuous and opportunistic BP monitoring without active user participation. Although promising, fully replacing cuff-based devices requires addressing multiple practical challenges, including user comfort, system stability, and regulatory and clinical validation.

VI. Conclusion

Contactless BP monitoring is a critical yet challenging problem in assessing health conditions, detecting potential diseases, and improving overall well-being. This paper serves as the first step towards such an ambition, driven by a sound physiological basis while leveraging commodity mmWave radar and interpretable DL technologies. We implement *h*BP-Fi based on a novel proposition to infer accurate BP via BP-encoded hemodynamics of pulses transiting along arm arteries. Specifically, we thoroughly study the hemodynamics and introduce BTF that characterizes the underlying BP information in pulse waveforms. To estimate BTF from RF signals, we design a beam-steerable sensing scheme that isolates pulse waveforms from multiple arterial sites, enabling recovery of transit waveforms beyond the radar's nominal angular resolution. Employing a carefully designed BTF-constrained CycleGAN, we equip *h*BP-Fi with an interpretable DL pipeline that ensures accurate BP estimation and effective adaptation to unseen subjects and scenarios. Experiments on 35 subjects across 12 scenarios show that *h*BP-Fi achieves promising cross-scenario BP estimation performance.

Acknowledgments

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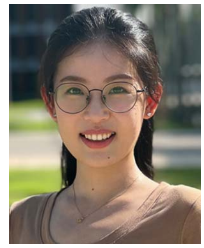

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